Exact Egyptian-fraction certificates and overlap-constrained search

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Abstract

We give explicit expansions of 1 into distinct unit fractions establishing $k(18) \leq 34$, $k(19) \leq 36$, $k(20) \leq 38$ and $k(21) \leq 39$. Each upper bound has a finite certificate verified by exact rational and integer arithmetic. An elementary cancellation lemma converts complete families of local shortening exclusions into overlap restrictions on every shorter representation. For the 34-term reference, complete five-to-four and fixed-core six-to-five searches give capacities 28 and 27. Classical quota optimization supplies exact reciprocal upper envelopes. A fixed 53-branch pilot compares three modes of one C++/GMP engine with a pinned original Sage implementation: the matched modes exhaust all 53 branches; Sage exhausts 48 and times out on five, with no contradictory completed answers. The existence of a 33-term expansion at $N = 18$ and the asymptotic question in Erdős problem 295 remain open.

1 Problem and contribution

For a positive integer N, let $k(N)$ be the least cardinality of a finite set D of distinct integers at least N satisfying

$$\sum_{d \in D} \frac{1}{d} = 1.$$

The main results are four explicit finite identities. Their term counts compare with the identified computational certificates as follows:

N	Cited certificate term count	Term count here
18	35	34
19	38	36
20	39	38
21	40	39

The $N = 18$ construction shortens the 35-term expansion recorded by White with Claude [2]. The comparisons for $N = 19, 20, 21$ use the vectors published by grind-35 on 24 September 2026 [3]. All source vectors used in these comparisons were rechecked with exact rational arithmetic. Each resulting expansion includes N itself, so it also gives an upper bound for OEIS A192881, whose definition requires the first denominator to equal N. The comparison is to the specified records.

Fixed-length enumeration and divisor-based two-term completion are classical [4]. Alekseyev provides related exact enumeration software in Sage [5]. Greedy optimization on matroids and laminar capacity families is established theory [6, 7]; membership-pattern dynamic programming and weak duality are standard methods. Here they accompany the explicit certificates, complete local-exclusion records and their elementary transfer into two concrete overlap restrictions. Finite allowed-denominator enumeration [8] is a related problem; the representation searches here allow unbounded denominators.

2 Four finite certificates

Proposition 1. *The following sets D_N have the stated cardinalities, consist of distinct integers at least N, and satisfy $\sum_{d\in D_N} 1/d = 1$.*

$N = 18$, 34 **terms:**

```
18,19,20,21,22,23,24,25,26,27,28,29,30,31,
32,33,34,35,36,38,39,40,43,44,45,46,
48,50,52,54,63,275,14143068,10457416992
```

$N = 19$, 36 **terms:**

```
19,20,21,22,23,24,25,26,27,28,29,30,31,32,
33,34,35,36,38,39,40,42,43,44,45,46,48,50,
52,54,55,66,70,275,14143068,10457416992
```

$N = 20$, 38 **terms:**

```
20,21,22,23,24,25,26,27,28,29,30,31,32,33,
34,35,36,37,38,39,40,41,42,43,44,45,46,
48,50,51,52,53,54,72,9512,316569706,
745735753538714400,755732875917998385
```

$N = 21$, 39 **terms:**

```
21,22,23,24,25,26,27,28,29,30,31,32,33,34,
35,36,37,38,39,40,41,42,44,45,46,48,49,50,
52,53,54,55,56,57,58,64,3162192,
13244735025984,60692489317197426150
```

Proof. Let L_N be the least common multiple of the displayed denominators. Exact integer addition gives the following certificates:

N	$L_N = \sum_{d\in D_N} L_N/d$
18	6209352774424800
19	6209352774424800
20	27729294399826116174306677448 00
21	207311948389939355504942400

Each denominator divides its stated L_N, and the integer quotients sum to L_N. Division by L_N proves the reciprocal identity. Cardinality, increasing order and minimum denominator are checked directly from the lists. □

The authoritative lists and all these integers are stored in `certificates-extension.json`, with large integers encoded as decimal strings. Verification of Proposition 1 is independent of the search that found the identities.

3 Construction and classical completion

For $N = 18$, retain 30 terms of the earlier 35-term certificate [2] and substitute

$$\frac{1}{42} + \frac{1}{640} + \frac{1}{337280} + \frac{1}{1735650} + \frac{1}{68479214496} = \frac{1}{46} + \frac{1}{275} + \frac{1}{14143068} + \frac{1}{10457416992}.$$

Multiplication by 636856694812800 gives 16160598373892 on each side. The four new denominators are distinct and absent from the retained terms. For $N = 19$, replace 18 and 63 in D_{18} by $42, 55, 66, 70$, using

$$\frac{1}{18} + \frac{1}{63} = \frac{1}{14} = \frac{1}{42} + \frac{1}{55} + \frac{1}{66} + \frac{1}{70}.$$

For $N = 20$ and $N = 21$, retain respectively the first 36 and 37 denominators of the cited 39- and 40-term certificates [3]. Replace their remaining three terms by the two-term completions

$$\frac{151817}{5698458937572121632800} = \frac{1}{745735753538714400} + \frac{1}{755732875917998385},$$
$$\frac{35558057}{470956940228352916800} = \frac{1}{13244735025984} + \frac{1}{60692489317197426150}.$$

The classical completion rule [4] follows from

$$(ax - b)(ay - b) = b^2. \tag{1}$$

For a positive reduced residual a/b and a lower endpoint D exceeding the retained denominators, a completion $a/b = 1/x + 1/y$, $D \le x < y$, exists exactly when a positive divisor q of b^2 satisfies

$$\max(1, aD - b) \le q < b, \qquad q \equiv -b \pmod{a}.$$

Then $x = (b + q)/a$ and $y = (b + b^2/q)/a$. A solution gives $q = ax - b$ by (1). Conversely, coprimality and the congruence imply that both denominators are integers; the inequalities give $D \le x < y$. The case $a = 1$ is immediate. Complete divisor enumeration decides completion for the specified prefix, rather than all prefixes. The reported $N = 20$ pair minimizes the larger denominator among completions of its fixed retained prefix, not globally.

4 An elementary transfer lemma

Let U be a universe, $w : U \to \mathbb{R}$ any weight function, and $W(S) = \sum_{x \in S} w(x)$ for finite S. Fix $A \subseteq U$ with $|A| = m$, integers $1 \le d \le r \le m$, and $C \subseteq A$ with $|C| \le r$. A target is a set B with $|B| = m - d$ and $W(B) = W(A)$.

For $R \subseteq A$, $|R| = r$, an admissible replacement is a set T such that

$$|T| = r - d, \qquad T \cap (A \setminus R) = \varnothing, \qquad W(T) = W(R).$$

Removed elements may reappear in T; only retained elements are forbidden. This convention is essential when padding the difference between two representations.

Lemma 2 (Fixed-core transfer)**.** *Suppose every r-element subset R of A containing C has no admissible replacement. Then every target B satisfies*

$$|B \cap (A \setminus C)| \le m - r - 1.$$

Proof. For a fixed R, a replacement produces the target $B = (A \setminus R) \cup T$ containing $A \setminus R$. Disjointness gives the cardinality and cancellation gives the weight. Conversely, a target containing $A \setminus R$ gives $T = B \setminus (A \setminus R)$ of cardinality $r - d$, disjoint from the retained set and of weight $W(R)$. Thus replacement existence is equivalent to the existence of a target containing the retained set.

Let $K = A \setminus B$. If $|K \cup C| \le r$, extend $K \cup C$ to an r-element subset $R \subseteq A$. Then $A \setminus R \subseteq B$, giving a replacement by the preceding equivalence, contrary to the premise. Hence $|K \cup C| \ge r + 1$, and complements inside A give

$$|B \cap (A \setminus C)| = m - |K \cup C| \le m - r - 1.$$

Neither positivity nor rationality of the weights is used. □

For $C = \varnothing$, every target overlaps A in at most $m - r - 1$ elements. This is an elementary cancellation lemma. The substantive computational premise is the complete family of exclusions.

5 Completed premises and reciprocal bounds

Take $A = D_{18}$ and $C = \{14143068, 10457416992\}$. The local searches use distinct denominators at least 18 and forbid exactly the retained terms:

Removed/added	Removed-subset family	Complete subsets	Seconds
5/4	Every 5-subset of A	$\binom{34}{5} = 278256$	95.781
6/5	Every 6-subset containing C	$\binom{32}{4} = 35960$	257.625

These families contain 314216 completed subsets. In the second family, the first log completes indices 0–33953 and interrupts index 33954. The resumed run replays 33954 and completes through 35959. The interrupted attempt is not counted as complete; the second time includes replay. `search-records/verify_local_premises.py` audits this coverage and boundary.

There is no denominator cap. At a positive reduced remainder a/b with $t \geq 2$ terms left and lower endpoint D, the next denominator lies in the finite necessary interval

$$\max(D, \lfloor b/a \rfloor + 1) \leq x \leq \lfloor tb/a \rfloor.$$

Exact harmonic bounds and proved arithmetic conditions prune branches. With two terms left, $y = bx/(ax-b)$ is checked exactly, without an imposed maximum on the final denominator. The full recurrence and arithmetic details are supplied in `proofs.md` and `SEARCH_STATUS.md`.

Lemma 2 gives, for every 33-term representation B with denominators at least 18,

$$|B \cap A| \leq 28, \qquad |B \cap (A \setminus C)| \leq 27. \tag{2}$$

These restrictions apply to the entire unbounded denominator universe. After a prefix P, subtract its selected overlap counts from the capacities. For t remaining denominators at least $D = \max(P) + 1$, accept denominators in increasing order when both residual capacities permit them. Their reciprocal sum is the exact maximum under these nested quotas, by classical greedy optimization for a truncated laminar matroid [6, 7]. A negative residual capacity, or a target above the maximum, excludes the branch.

For crossing finite quota groups, partition denominators by membership pattern. Within each pattern and chosen count, use the smallest denominators; dynamic programming over counts gives the exact upper envelope. Only the first t members of each pattern are needed. This finite support evaluates a bound and does not restrict the representation search. Nonnegative exact multipliers give standard weak-duality bounds. Proofs, resource guards and executable checks are supplied in `STRUCTURE.md` and `structure/`.

6 A controlled original-Sage pilot

The frozen corpus has 224 prefixes, each extended toward a 33-term representation of 1 with minimum denominator 18:

- 190 broad prefixes $\{18, \ldots, 28\} \cup \{30, \ldots, 44\} \cup \{u, v\}$, $45 \leq u < v \leq 64$;
- 31 structural prefixes: the first 27 elements of A, followed by $z = 49, \ldots, 79$;
- three controls $\{18, \ldots, 28\} \cup \{z\}$ for $z = 31, 32, 33$.

The pilot includes all structural prefixes and controls, plus every tenth broad prefix in lexicographic (u, v) order, starting at ordinal zero: 53 cases fixed before timing.

The pilot used one Linux runner CPU and one repetition per case. The unchanged original Sage source [5] was fetched at commit `8680599e77683a9057ee55871a0893bb319300e8`, with the recorded SHA-256 checked before execution. It ran in SageMath 10.8 with one map-reduce worker. The wrapper called `one_rep` on each exact residual with `distinct=True`, the correct lower endpoint and no maximum denominator. Six API controls passed, including the exact set of all six four-term distinct expansions of 1 with denominators at least 2.

One C++/GMP source ran three modes: OFF, full-reference capacity 28, and nested capacities 28/27. Arithmetic, traversal, prime/quadratic pruning and memo limits were shared. Ordinary cases had a five-second solver budget and the frontier controls had 30 seconds:

Solver/mode	Exhausted	Timed out	Errors
Original Sage	48	5	0
Matched GMP, OFF	53	0	0
Matched GMP, single quota	53	0	0
Matched GMP, nested quotas	53	0	0

All 48 branches completed by both programs agreed on absence. No contradictory completed answer occurred. The five Sage timeouts remain censored baseline runs. The completed matched searches exclude these specified branches, rather than all 33-term possibilities. Saved records, source/image hashes and the completed workflow are available in [9].

The inter-program comparison includes C++/GMP versus Sage/Python execution, different tail routines, harmonic/prime/quadratic pruning and worker-management costs. Its timing difference does not isolate the effect of overlap quotas. The three matched modes provide that ablation. For example, `structural-052` visits 9444, 7276 and 1 states in OFF, single and nested modes, respectively, with the same exhausted verdict. This demonstrates a concrete quota exclusion in the selected family. A single pilot and selected structural branches do not establish a general runtime advantage.

Solver timers exclude compilation, Sage startup/source loading and deriving the local-exclusion premises. Sage API timing includes worker setup and cleanup; separate wall times are retained. The historical premise times were measured separately and are not hardware-normalized to the cloud pilot. Timeouts and late answers remain in the records.

Independent unpruned rational enumeration checked 972 conditional residual states with one, two or three terms left: 349 admitted a witness and 623 were exhausted. Verdicts agreed with the matched engine, and every returned witness passed exact sum, order, distinctness and quota checks. These are implementation and boundary controls, not a global 33-term exclusion.

6.1 Repeated local quota ablation

A separate Windows run executed all 224 cases in all three matched modes, with three repetitions: 2016 solver runs. The declared budget was two seconds per ordinary case and 30 seconds per control. Strict post-processing censored nine raw timeouts and four answers returned after their deadline; all 2016 records are retained. There were no errors or contradictory completed verdicts. Exactly 213 cases finished within budget in all nine runs.

For each case and mode, take the median of its three solver times. The table reports the sum of these medians for OFF divided by the sum for nested quotas, restricted in each row to cases whose nine runs all completed within budget. A ratio above 1 favors nested quotas on this conditional measure.

Family	Complete / declared	OFF / nested
All	213/224	1.05244
Broad	181/190	1.01684
Structural	29/31	1.67216
Controls	3/3	0.92689

The overall difference is modest, with the larger effect concentrated in the selected structural family. The broad family changes little, while controls take longer with nested quotas on this measure. These ratios concern online solver time on a common completed subset; they exclude premise computation and compilation, and do not measure the full suite including censored runs. For the same 213 instances, sums of median external wall times were 108.386984 seconds for OFF and 109.278968 for nested quotas. Process launch, output collection and validation therefore removed the small aggregate internal-time gain in this execution. The Windows repetitions and the Linux Sage pilot are separate experiments.

7 Reproducibility and remaining questions

Code and records are available in [10]. Python 3.9+ and its standard library suffice for `python verify-extension.py`; a separately written BigInt check runs as `node verify-extension-bigint.mjs`.

The verifier checks rational sums, integer LCM identities, cardinality, order, minima and source-to-result substitutions. Twenty deliberately corrupted records are rejected. The general transfer and envelope checks run as `python structure/general_theorem_oracle.py`. Local-exclusion audits and search programs are preserved separately; search is unnecessary to verify Proposition 1.

The accompanying release reproduces the previously reported exclusion of exactly 32 terms at $N = 18$, using the same mathematical recurrence in separately written exact-arithmetic implementations. The logged run has 814918 visited states; all two-term states are rejected before candidate enumeration. Excluding 32 terms also excludes every shorter expansion: repeatedly replace its largest denominator d by $d+1$ and $d(d+1)$, using

$$\frac{1}{d} = \frac{1}{d+1} + \frac{1}{d(d+1)},$$

until 32 terms are reached. The new denominators exceed every retained one. Thus the recorded bounds are $33 \le k(18) \le 34$. Whether $k(18)$ is 33 or 34 remains unresolved here.

Erdős problem 295 asks whether $k(N)-(e-1)N \to +\infty$ [1]. These finite certificates and overlap exclusions do not settle that question. Further progress would require a completed minimum proof, a quantitatively stronger family of constructions or a broader controlled study of the pruning rules.

Computational contribution. Kyunghan Bae initiated the project; OpenAI Codex agents implemented and ran the searches, found the certificates, and assisted with verification and exposition.

References

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